PHAT-JeT: Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging

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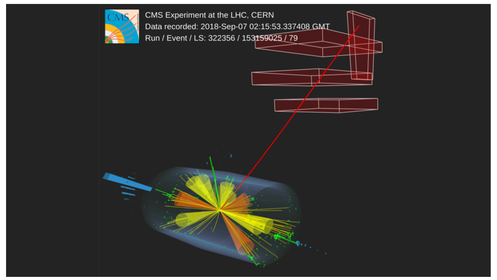
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javierzhao.github.io/PHAT-JeT

1. Jet Tagging at the LHC Trigger

- **40M collisions/s**, raw data **> 1 PB/s**. Most events are known background; only a small fraction can be stored.
- The FPGA Level-1 trigger decides within $O(10\ \mu s)$ which events to keep; jet tagging gets **< 100 ns** (~1M FLOPs per jet).
- **Jet tagging:** identify the particle behind a jet (quark, gluon, W, Z, top) from up to 150 unordered particles.
- Small accuracy gains mean more rare signal events recorded.



CMS event display: jets (yellow/orange cones) from one collision.

- No previous attention-based method achieves high accuracy and also fits within the strict trigger budget of **100 ns**.
- Low-rank, linear and hashing variants fit, but drop exact pairwise interactions and lose accuracy.

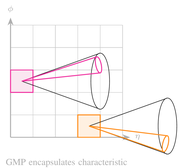
2. Related Methods

Method	Exact attention	Trigger-scale
Set Transformer	✗	✗
PaCa-ViT	✗	✗
Swin	✓	✗
ParT	✓	✗
JEDI-Linear	✗	✓
PHAT-JeT	✓	✓

PHAT-JeT is the only method with exact attention at trigger scale.

- **Linformer:** low-rank projection drops pairwise terms.
- **JEDI-Linear:** linear-time, no explicit pairwise edges.
- **HEPT:** hashing is unstable at small patch sizes.
- **Swin, Longformer:** need a grid or sequence order.

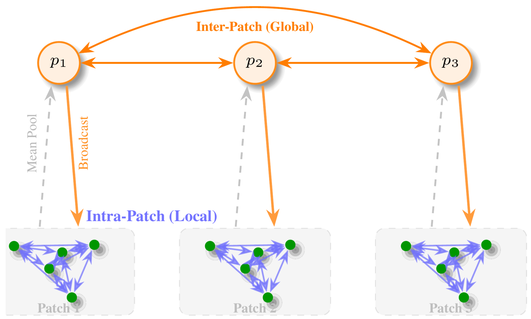
3. PHAT-JeT Architecture



1 Geometric Message Passing

Particles are binned on a coarse (η, ϕ) grid. A depthwise 2D convolution mixes neighboring cells and adds the result back to each particle, encoding multi-prong structure.

$$G_{u,v} = \sum_{i:(u_i,v_i)=(u,v)} X_i, \qquad X_i \leftarrow X_i + W_p\,\mathrm{DWConv}(G)_{u_i,v_i}$$



2 Intra-patch (local)

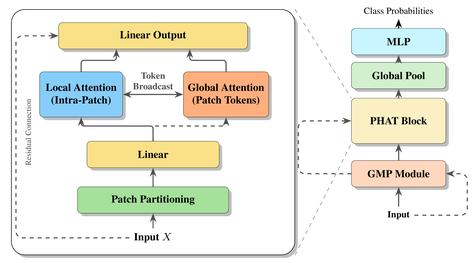
Exact self-attention inside each patch of P particles. Cost $O(NP)$, not $O(N^2)$.

$$Z^{(j)} = \mathrm{Attn}\big(Q^{(j)}, K^{(j)}, V^{(j)}\big)$$
$$Q^{(j)} = X^{(j)} W_Q, \text{ etc.}$$

3 Inter-patch (global)

Each patch is averaged into a token p_j; tokens attend to each other and are added back to their patch.

$$p_j = \tfrac{1}{P}\textstyle\sum_i x_i^{(j)}$$
$$\tilde{p} = \mathrm{MHSA}(p_1, \ldots, p_{N/P})$$
$$x_i^{(j)} \leftarrow x_i^{(j)} + W_m\,\tilde{p}_j$$



Full model: input → GMP 1 → PHAT block (local 2 + global 3) → pool → MLP.

4. Datasets

hls4ml
CMS Level-1 trigger emulation; 5 classes.

JetClass
10 classes; trained on a 2M-jet subset.

Top Tagging
Top vs. light-quark/gluon jets.

Quark–Gluon
Quark vs. gluon jets.

Background rejection = 1 / (false-positive rate) at 80% signal efficiency. Higher is better; the main trigger metric.

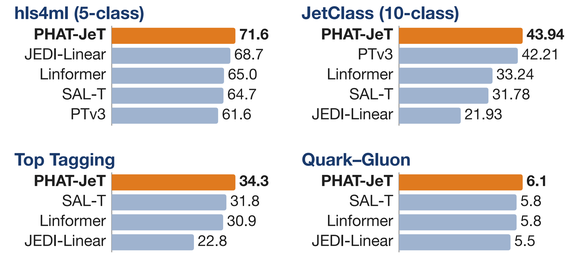
5. JetClass Results

128 particles per jet; efficient models at ~1.2–1.5M FLOPs.

Model	Acc (%)	Bkg Rej	Params	FLOPs
PHAT-JeT	**65.38 ± 0.08**	**43.94 ± 0.65**	**6.8K**	**1.24M**
Point Transformer V3	64.69 ± 0.11	42.21 ± 0.25	23.7K	1.45M
Linformer	62.64 ± 0.19	33.24 ± 0.11	9.5K	1.49M
SAL-T	62.19 ± 0.28	31.78 ± 0.90	7.0K	1.42M
JEDI-Linear	59.52 ± 0.82	21.93 ± 1.64	6.1K	1.31M
Small ParT	57.58 ± 0.28	15.83 ± 0.43	2.9K	1.52M
Transformer	56.18 ± 1.34	15.09 ± 2.53	1.0K	1.24M
PointNet	55.87 ± 0.16	20.83 ± 0.70	9.9K	1.35M
Unconstrained references (10^2–10^4× more FLOPs)				
ParT (full)	*83.6*	*2345.6*	*2,140K*	*340M*
L-GATr	*83.9*	*2844.2*	*1,682K*	*9,709M*

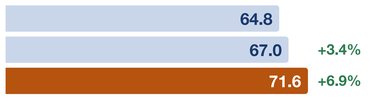
Best accuracy and background rejection at matched FLOPs: +0.69 pts over PTv3 with 3.5× fewer parameters, ≥2.7 pts over the rest.

6. Background Rejection by Benchmark



PHAT-JeT has the highest background rejection on all four benchmarks.

7. Ablation (hls4ml bkg rej)



Accuracy: 81.23 → 81.51 → 81.80%.

8. Beyond HEP: ScanObjectNN Results

Real 3D object scans (PB_T50_RS, 1024 points); same block, 3D voxel grid for GMP.

Method	Params (M)	OA (%)
PointNet++	1.5	77.9
SimpleView	0.8	80.5 ± 0.3
PHAT-JeT-XS	**0.157**	**81.22 ± 0.16**
PHAT-JeT-S	**0.612**	**82.29 ± 0.41**
PointMLP-elite	0.68	83.8 ± 0.6
PointNeXt-S	1.4	87.7 ± 0.4

PHAT-JeT-XS beats PointNet++ and SimpleView with 5–10× fewer parameters; larger point-cloud models still lead.

Selected rows; baselines from published tables.

9. FPGA Estimates

AMD VU13P, 300 MHz, 64 particles; target **< 100 ns**; pre-route estimates.

Design	Bkg Rej	LUTs	ns
JEDI-Linear, official	66.0	69.9k	63.3
JEDI-Linear, official perm-invariant	80.4	158k	73.3
JEDI + GMP	84.8	144k	69.9
JEDI + GMP + patch tokens	84.0	208k	96.6
Patch attn (ReLU) + GMP	66.7	708k	99.9

Each PHAT-JeT component has a variant under 100 ns. GMP gives the highest background rejection (84.8); exact patch attention fits only with ReLU and 708k LUTs.

Official JEDI-Linear designs use 64 particles; its performance degrades when extended to 128.

TL;DR: exact attention within patches and across patch tokens; best performance among trigger-scale models on four benchmarks.

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